The triple constraint (scope, time, cost)

Every project balances scope, time and cost, with quality as the outcome of that balance; changing one side changes the others. This chapter teaches you to recognise the constraints, negotiate trade-offs and communicate them with numbers.

PMBOK Guide 7th ed. | Agile Practice Guide 2017 | Case: Realtime Quiz

AGENDA

What we will cover

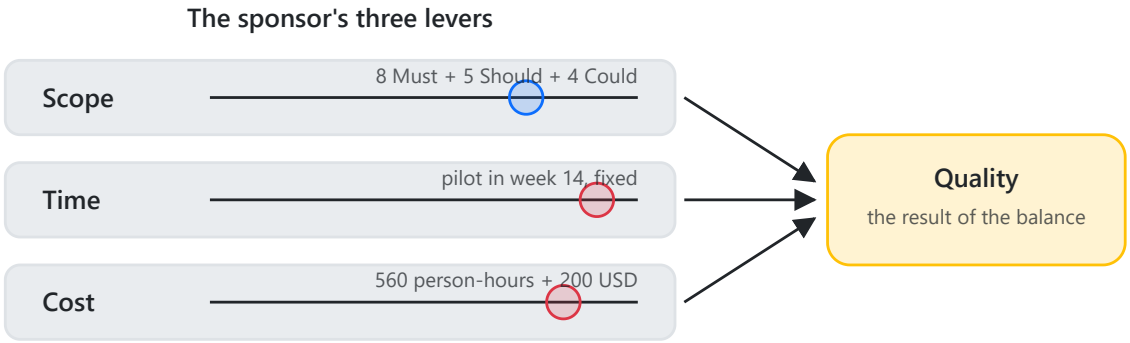
1. The iron triangle and its extended forms (quality, risk, resources, benefits)
2. Scope creep, gold plating and how they consume time and cost
3. Fixed-date, fixed-budget and fixed-scope projects and which side flexes
4. Trade-off analysis and prioritisation (MoSCoW, weighted scoring)
5. Crashing and fast-tracking as time-for-cost trades
6. Constraints versus assumptions and how to document them
7. Measuring the constraints: baselines and variances
8. The constraint model in agile projects: fixed time and cost, variable scope
9. Negotiating with sponsors and saying no with data
10. Case discussion: an IT project that traded scope for a deadline
11. Chapter Quiz
12. Lab-02

Click any item to jump directly to that topic.

INTRODUCTION

Three levers, one result: why every project is a negotiation

- Chapter 01 gave us the **charter** of the Realtime Quiz project: a pilot with 50 students in **week 14**, a team of **4 students plus a lecturer** as project manager, and a cash budget of about **200 USD**.
- Those three facts are the levers a sponsor can pull: **scope** (what we deliver), **time** (when) and **cost** (with what effort and money).
- You cannot pull one lever without moving the others. **Quality** is what comes out when the three are balanced, or what breaks when they are not.
- The job of a project manager is not to say yes or no. It is to show, with numbers, what each choice costs, and to let the sponsor choose.
- Lab-02 turns this into a constraint analysis and a one-page trade-off memo for the Realtime Quiz sponsor.



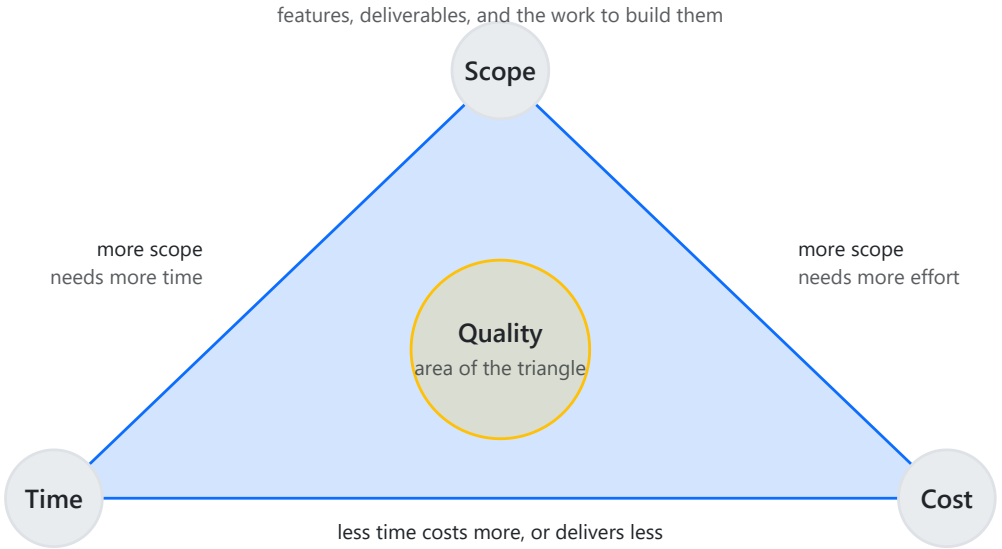
red knobs are fixed by the charter: only the blue one can move freely

TOPIC 1

The iron triangle: scope, time, cost, and quality in the middle

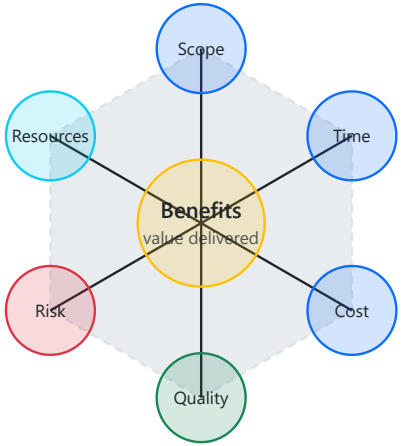
- **Scope**: the features and deliverables, including the work needed to produce them.
- **Time**: the schedule, milestones and the end date.
- **Cost**: effort (person-hours), money, equipment and licences.
- **Quality** sits in the centre: it is the area of the triangle. Squeeze one side without adjusting the others and the area shrinks: more defects, fewer tests, unhappy users.
- The model dates from the 1950s (Martin Barnes). The PMBOK Guide 7th ed. no longer draws it but keeps the idea: constraints are competing factors that limit the team's options.

> 💡 **Rule of thumb**: "Good, fast, cheap: pick two." The third one is where the project gives way.



TOPIC 1

Beyond three: the six-constraint star and the benefits in the middle



Pull one point, the others move

Cut time: risk goes up, cost goes up, or scope and quality come down.

Add resources: cost up, and time down only if the work can be split.

Accept more risk: cheaper and faster now, expensive when the risk fires.

The centre is why the project exists: a project on time and on budget that delivers no benefit has still failed.

PMBOK Guide 7th ed.: value delivery, not triangle

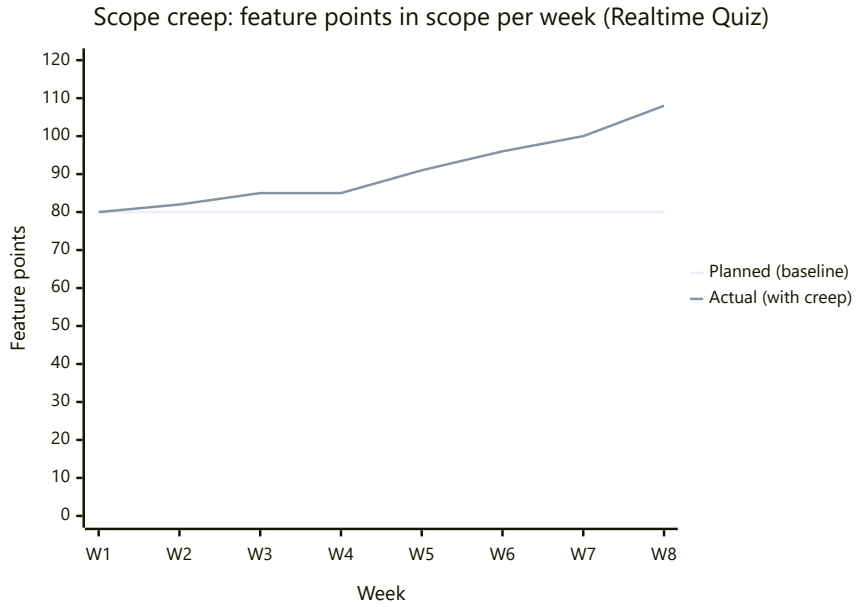
Constraint	Realtime Quiz value (from the charter)	Set by
Scope	8 Must features (MVP), 5 Should, 4 Could; 47 functional requirements	Software Engineering plan
Time	14-week semester, pilot with 50 students in week 14, MVP checkpoint end of week 7	Academic calendar
Cost	4 students × 10 h/week × 14 weeks = 560 person-hours; cash budget 200 USD (VPS 25 USD/month, domain, tools)	Dean, sponsor
Quality	NFRs: broadcast under 500 ms, WCAG AA, tests green in CI, 20 real participants in the demo	Definition of Done
Resources	Linode VPS 2 vCPU / 2 GB, student laptops, campus network	IT department
Risk	New Spring Boot 4 stack, WebSocket blocked by proxy, scope creep from Could features	Risk register (Lab-09)
Benefits	Live quizzes in every course, students pass the capstone, replaces a paid tool	Business case

⚠ Do not treat quality as a lever. Sponsors rarely ask for lower quality; it drops silently when the other three are over-constrained.

TOPIC 2

Scope creep and gold plating: how scope grows when nobody decides

- **Scope creep**: uncontrolled growth of scope through small, unapproved additions. Each one "only takes a day"; together they eat the contingency and then the deadline.
- **Gold plating**: the team adds features nobody asked for (a dark mode, an animation, a fifth chart) because they are fun or "obviously useful". It costs time and creates untested code.
- Both are **invisible on the plan**: the baseline still says 80 points, the work says 108. That is a 35 % overrun with no approval on record.
- The cure is not to refuse change but to make every change **visible and priced**: a change log, an estimate, a decision by the sponsor.
- Realtime Quiz plan, risk table: "Scope creep from Could features: freeze scope at the MVP checkpoint."



Planned 80 points; actual 108 by week 8: (108 - 80) / 80 = 35 % more scope, with unchanged time and cost.

A scope change log: three requests, three decisions

```
SCOPE CHANGE LOG · Realtime Quiz · owner: PM (lecturer)
Baseline: 504 h planned work + 56 h contingency = 560 h capacity

ID     Week  Request (who asked)              Est.   Decision
CR-01  W3    Chat during sessions (sponsor)   40 h   Defer: stays Could,
                                                     revisit at MVP demo
CR-02  W4    Anonymous join with nickname     12 h   Accept: swap for the
             (student representative)                PDF export (24 h)
CR-03  W5    Dark mode on the play page (team) 6 h   Reject: gold plating,
                                                     not in any FR

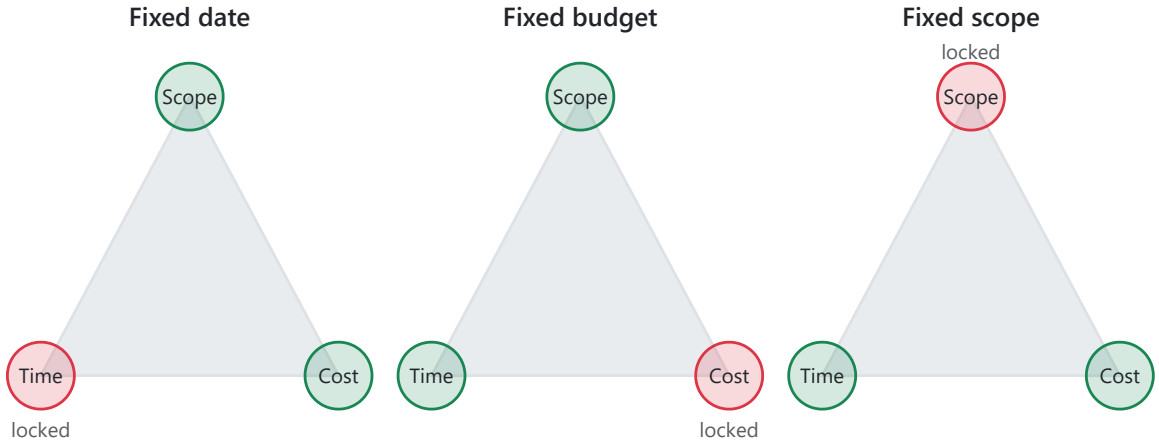
Net effect on baseline: +12 h - 24 h = -12 h  (contingency untouched)
Rule: nothing enters the sprint backlog without a row in this log.
```

Column	Why it is there
Week	Late requests cost more: the same 12 h in week 12 would hit the pilot.
Who asked	Sponsor requests go to the sponsor for a trade; team ideas are gold plating until a user asks.
Est.	An estimate turns "just a small thing" into hours the sponsor can weigh.
Decision	Accept, defer, reject, or swap: and always what was given up in return.

ⓘ **Common mistake**: accepting CR-02 without the swap. Twelve hours here, six there, and the 56 h contingency is gone before the riskiest phase (live sessions, weeks 5-6) starts.

TOPIC 3

Fixed-date, fixed-budget, fixed-scope: decide early which side may move



conference demo, tax law change, semester pilot in week 14
cut scope or add cost

grant-funded research tool, fixed-price contract
trim scope or take longer

regulatory compliance, safety system, data migration (all records or none)
pay more or take longer

- Ask the sponsor in the charter phase: **"If something has to give, what gives first?"** The answer is the project's driver; the other two sides are the buffers.
- **Fixed date**: the calendar cannot move. Flex scope (deliver less) or cost (add people, buy instead of build).
- **Fixed budget**: the money cannot grow. Flex scope or take longer with the same team.
- **Fixed scope**: every requirement is mandatory. Flex time or cost; quality is often the hidden victim here.
- Two sides fixed at once is common and workable. **Three sides fixed is a wish, not a plan**: the team will fix the fourth thing, quality, without telling you.

Classifying the Realtime Quiz project from its charter

```
Constraint  Value in the charter            Can it move?   Verdict
Time        Pilot with 50 students in W14;  No: semester   FIXED
            grades due at semester end      ends, grades   (driver)
Cost        4 students x 10 h/week;         No: no more    FIXED
            200 USD cash                    students, no
                                            more cash
Scope       8 Must + 5 Should + 4 Could     Yes: Should    FLEXES
                                            and Could are
                                            negotiable
Quality     NFRs and Definition of Done     Must not move  PROTECTED

Consequence: the project is fixed-date AND fixed-budget.
Only scope can absorb surprises. Every Should and Could feature
is a buffer; every Must feature that grows must push a Should out.
```

Project type	Example	Driver	Flexes
Student capstone	Realtime Quiz	Date and budget	Scope (Should, Could)
Event system	Conference registration site	Date	Scope, then cost
Compliance	New tax rate in the invoicing module	Scope and date	Cost (overtime, contractors)
Open-source library	Volunteer-run framework release	Scope and cost	Time ("it ships when ready")
Public tender	Fixed-price ERP roll-out	Cost and scope	Time, and quality if unmanaged

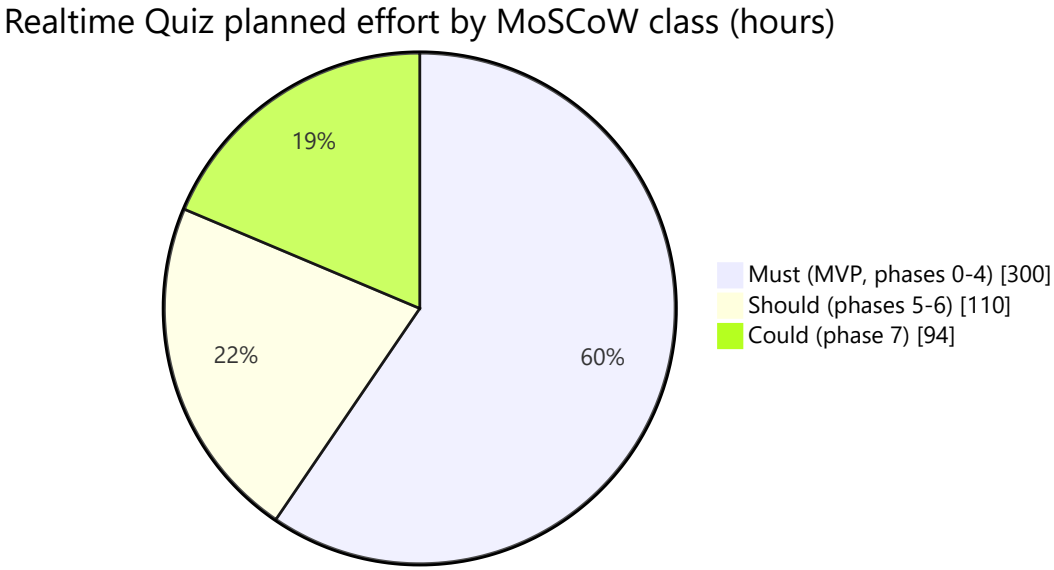
Evidence, not opinion: in Lab-02 Task 1 you quote the charter sentence that makes each constraint fixed. "The pilot is in week 14" is evidence; "the sponsor probably wants it soon" is not.

TOPIC 4

MoSCoW: sorting scope into what can flex

- **Must**: without it the release is useless or illegal. Test: "if this is missing, do we cancel the pilot?"
- **Should**: important, painful to leave out, but there is a workaround for one release.
- **Could**: nice to have; first to go when time runs short.
- **Won't (this time)**: explicitly out of scope now, written down so nobody re-argues it.
- The DSDM guidance: keep Must at or below **60 % of the effort**; the remaining 40 % is the buffer that makes a fixed date achievable.

Class	Features	Effort	Share
Must	8	300 h	60 %
Should	5	110 h	22 %
Could	4	94 h	18 %
Won't	native apps, LMS, free text	0 h	0 %
Total	**17**	**504 h**	**100 %**



Must = 300 / 504 = 59.5 %: just inside the 60 % guideline. If Must grows to 340 h (67 %), the fixed date is at risk.

Weighted scoring: choosing between options with numbers

```
Decision: how do we get a quiz platform for the week-14 pilot?
Scores 1 (poor) to 5 (excellent); weights sum to 100.

Criterion                 Weight   Build    Buy SaaS   Hybrid
Learning value (capstone)   40       5         1          3
Cost within 200 USD         25       4         3          3
Time to week-14 pilot       20       3         5          4
Fit to requirements         15       5         3          4

Weighted score = sum(weight x score)
Build  = 40*5 + 25*4 + 20*3 + 15*5 = 200 + 100 + 60 + 75 = 435
Buy    = 40*1 + 25*3 + 20*5 + 15*3 =  40 +  75 + 100 + 45 = 260
Hybrid = 40*3 + 25*3 + 20*4 + 15*4 = 120 +  75 +  80 + 60 = 335

Maximum = 100 x 5 = 500  ->  Build 87 %, Hybrid 67 %, Buy 52 %
Sensitivity: drop "Learning value" to 20 and raise "Time" to 40:
Build = 100+100+120+75 = 395, Buy = 20+75+200+45 = 340. Build still wins.
```

Spreadsheet layout (Lab-02 Task 3): weights in column B, one column per option, formulas written out.

```
   A                B       C      D      E
1  Criterion        Weight  Build  Buy    Hybrid
2  Learning value   40      5      1      3
3  Cost             25      4      3      3
4  Time to pilot    20      3      5      4
5  Fit              15      5      3      4
6  Weight check     =SUM(B2:B5)          -> 100
7  Score            =SUMPRODUCT($B$2:$B$5,C2:C5)
   (copy C7 to D7 and E7)
8  Percent          =C7/(5*$B$6)
```

⚠ Agree on the **weights before** you score the options. Weights chosen after the scores are known always favour the option someone already wanted.

A trade-off memo: adding chat to the Must scope without moving the pilot

```
TRADE-OFF MEMO · CR-01 "Add chat to the Must scope"          Week 3
To: Head of IT (sponsor)    From: PM (lecturer)    Decision needed by: W4
Fixed: pilot with 50 students in week 14; team 4 x 10 h/week; 200 USD
Request: session chat (FR-N3), estimate 40 h, currently a Could feature

Option                              Time        Cost        Quality
A  Add chat, keep every Should      +1 week     0 USD       pilot slips to W15
B  Add chat, defer PDF export       +0 weeks    0 USD       CSV export only
   (24 h) and video media (16 h)                            at the pilot
C  Add chat, hire a TA for 40 h     +0 weeks    +480 USD    over budget by
   at 12 USD/h                                              280 USD
D  Text-only chat (16 h), no mute,  +0 weeks    0 USD       contingency 56 h
   paid from contingency                                    drops to 40 h

Recommendation: B. A breaks the fixed date, C breaks the budget, and
D spends contingency before the riskiest phase (live sessions, W5-W6).
If B is approved I re-baseline scope today; total effort stays 504 h.
```

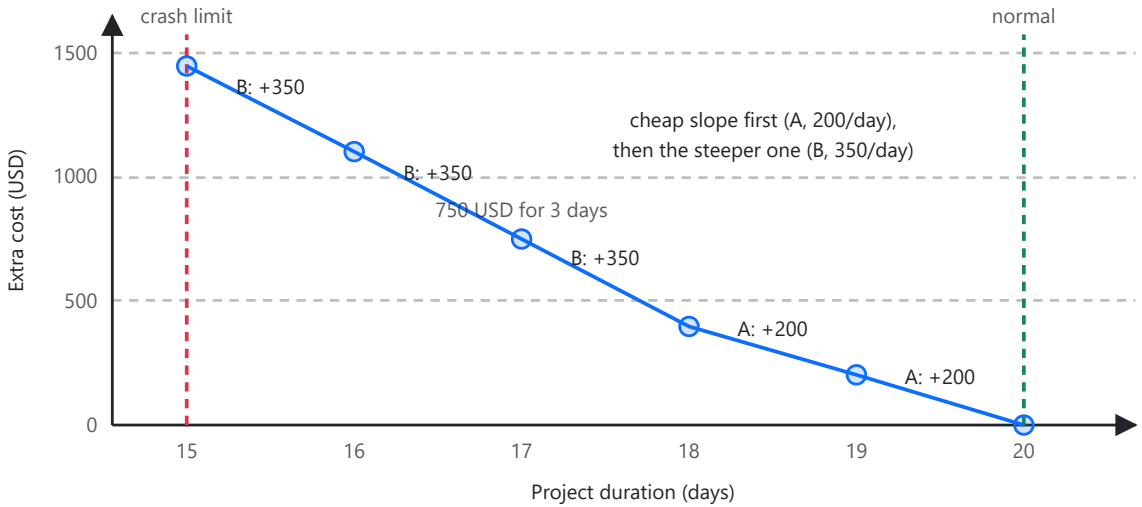
Memo part	Purpose
Header	Who decides, and by when: a memo without a deadline is a wish list.
Fixed facts	Remind the sponsor which sides cannot move and where that was agreed.
Options table	At least three options, each priced in time, cost and quality. Never one option.
Recommendation	One choice, one sentence of reasoning per rejected option.

💡 The memo never says "no". It says "yes, and here is the price of each yes". The sponsor keeps the decision; the PM keeps the numbers honest.

TOPIC 5

Crashing and fast-tracking: buying time with money or with risk

- **Schedule compression** (PMBOK Guide 7th ed.) shortens the schedule without cutting scope. Two techniques.
- **Crashing**: add resources to critical-path activities (overtime, a contractor, a faster server). Trades **cost for time**. Each activity has a crash limit and a cost per day saved.
- **Fast-tracking**: run activities in parallel that were planned in sequence (start the play page before the WebSocket API is final). Trades **risk and rework for time**; costs little cash.
- Only activities on the **critical path** shorten the project. Crashing a non-critical task buys nothing.
- The crash cost curve bends upward: the first day is cheap, the last day is expensive. Stop when the next day saved costs more than a day of delay.



Which activity to crash first: A at 200 USD per day or B at 350?

```
Situation: the MVP checkpoint is 3 days late. A (WebSocket session
core) and B (scoring + leaderboard) are both on the critical path.

Activity  Normal   Crashed  Normal cost  Crash cost  Cost/day  Max days
A         6 days   4 days    800 USD     1200 USD    200 USD   2
B         8 days   5 days   1400 USD     2450 USD    350 USD   3

Cost per day saved = (crash cost - normal cost) / (normal - crashed)
A: (1200 - 800) / (6 - 4)  =  400 / 2 = 200 USD per day
B: (2450 - 1400) / (8 - 5) = 1050 / 3 = 350 USD per day

Step  Action              Extra   Running total   Why
1     crash A by 1 day    +200    200             cheapest slope
2     crash A by 1 day    +200    400             A reaches its limit
3     crash B by 1 day    +350    750             next cheapest

Alternative: crash B alone by 3 days = 3 x 350 = 1050 USD (300 more).
After each step re-check the critical path: shortening A or B may make
another path (for example the front-end pages) critical instead.
```

	Crashing	Fast-tracking
Pays with	Money, effort	Risk, rework
Works when	Work can be split among more people	Dependencies are soft (a draft API is enough to start)
Fails when	Brooks's law: adding people to a late software project makes it later	The upstream design changes and downstream work is redone
Realtime Quiz	Paid TA hours; a bigger VPS for load tests	Build the play page against a mocked STOMP topic while the server is unfinished

⚠ For a student team with no cash for crashing, fast-tracking is the realistic tool. Price its risk: "if the WebSocket event names change, the play page costs 8 h of rework."

TOPIC 6

Constraints versus assumptions: what is imposed and what is believed

	Constraint	Assumption
Definition	A limiting factor imposed on the project that the team cannot change (PMBOK Guide 7th ed.)	A factor considered true for planning purposes, without proof
Who sets it	Sponsor, law, calendar, contract, technology decision	The team, when it has to plan with incomplete information
What can happen	Violating it fails the project or breaks an agreement	It turns out false and becomes a risk or an issue
Action	Plan inside it; escalate if it makes the goal impossible	Validate by a date; name an owner; link to the risk register
Test	"Can we decide to ignore it?" No.	"Could we be wrong about it?" Yes.

💡 Same sentence, different type: "The VPS costs 25 USD per month" is a *constraint* if the sponsor capped the budget at that, an *assumption* if the team looked up a price list and hopes it holds.

The log template (Lab-02 Task 5). Assumptions carry a validation date; constraints carry the source that imposed them.

```
CONSTRAINTS AND ASSUMPTIONS LOG · <project> · v<n> · <date>

ID    Type        Statement          Source or owner   Validate by
C-nn  Constraint  what is imposed    who imposed it    n/a (fixed)
A-nn  Assumption  what we believe    who checks it     week / date
                                                       -> risk R-nn

Rules
- One statement per row, measurable ("50 students", not "many").
- Every assumption has an owner and a date; overdue = open issue.
- A false assumption becomes a risk (Lab-09) or a change request.
- Review the log at every milestone; retire validated assumptions.
```

The Realtime Quiz constraints and assumptions log, week 2

CONSTRAINTS AND ASSUMPTIONS LOG · Realtime Quiz · v0.2 · week 2

ID	Type	Statement	Source / owner	Validate
C-01	Constraint	Pilot with 50 students in week 14	Charter, sponsor	fixed
C-02	Constraint	Team: 4 students x 10 h/week + PM	Charter, dean	fixed
C-03	Constraint	Cash budget 200 USD (VPS, domain)	Charter, sponsor	fixed
C-04	Constraint	Stack: Spring Boot 4.1, PostgreSQL, Bootstrap 5, no SPA framework	SE course plan	fixed
C-05	Constraint	Passwords hashed, no secrets in git	Security policy	fixed
A-01	Assumption	Campus network allows WebSocket (fallback: SockJS XHR)	Team lead	W5 -> R-04
A-02	Assumption	A 2 vCPU VPS serves 50 players with broadcast under 500 ms	Backend dev	W11 -> R-06
A-03	Assumption	Each student really has 10 h/week (exam weeks 7 and 13 excepted)	PM	W4 -> R-02
A-04	Assumption	50 volunteers can be recruited for the pilot	Sponsor	W10 -> R-09
A-05	Assumption	Noto Sans Khmer renders in the PDF export library	Frontend dev	W8 -> R-07

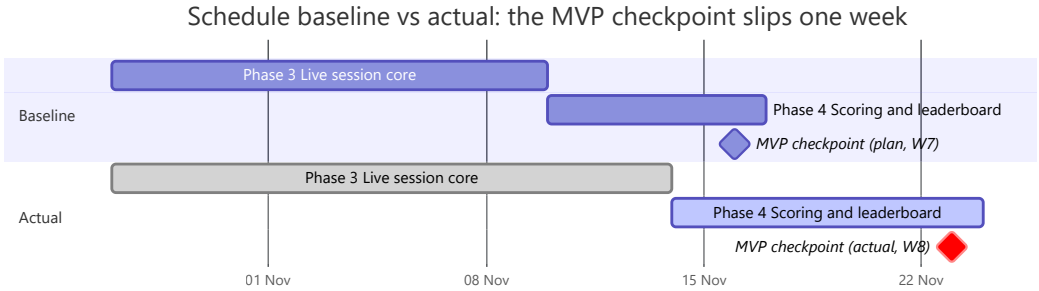
- **C-01 to C-03** are the three sides of the triangle. C-04 and C-05 are technical constraints: imposed by the Software Engineering course, not negotiable in this project.
- **A-01** is the classic hidden killer: if the campus proxy blocks WebSocket, the whole Must scope is at risk. Validate it in week 5, before the live-session phase starts, not in week 14.
- **A-03** is an assumption about cost. If exam weeks cut capacity by 40 h, the fixed-budget side has just moved without anyone deciding.
- Each assumption links to a risk id (**R-nn**); Lab-09 will build the register from these rows.

ⓘ **Common mistake**: listing "the team is motivated" as an assumption. It is not measurable and cannot be validated. Write "each student logs at least 8 h/week in the timesheet by W4".

TOPIC 7

Measuring the constraints: baselines, variances, and a slipped milestone

- A **baseline** is the approved version of a plan: the **scope baseline** (scope statement, WBS), the **schedule baseline** (milestone dates), the **cost baseline** (time-phased budget). Together they are the *performance measurement baseline*.
- A baseline changes only through an approved change request. Otherwise you cannot tell a plan that slipped from a plan that was rewritten.
- **Variance** = actual - baseline. Report it in the unit of the constraint: weeks, hours, USD, features.
- Variance is a signal, not a verdict. Ask: is it a one-off, or a trend that forecasts the end date?
- Chapter 06 adds earned value (SV, CV, SPI, CPI): the same idea with money as the common unit.



Phase 3 took 18 days instead of 14 (WebSocket harder than estimated); phase 4 inherited the delay. Schedule variance at the milestone: -1 week.

A variance report at week 8 and what it forecasts

```
VARIANCE REPORT · Realtime Quiz · status date: end of week 8

Measure                 Baseline          Actual            Variance
MVP checkpoint          W7 (2026-11-16)   W8 (2026-11-23)   -1 week
Effort to MVP           280 h             310 h             +30 h
Must features done      8 of 8            7 of 8            -1 feature
Cash spent              60 USD            60 USD            0 USD

Variance % = (Actual - Baseline) / Baseline x 100
Effort:   (310 - 280) / 280 x 100 = 10.7 % over plan
Schedule: 1 week late on a 7-week baseline = 14.3 % late

Trend: 30 h lost over 7 weeks = 4.3 h per week
Forecast at week 14 if the trend continues: 4.3 x 14 = 60 h over,
against a contingency of 56 h  ->  the buffer is gone by week 13.

Actions proposed to the sponsor (fixed date, fixed budget):
1. Move "video in questions" (16 h) and "question bank" (30 h) to Won't.
2. Fast-track the results page against mocked data (risk: 6 h rework).
3. Re-baseline the milestones after approval; keep the pilot at W14.
```

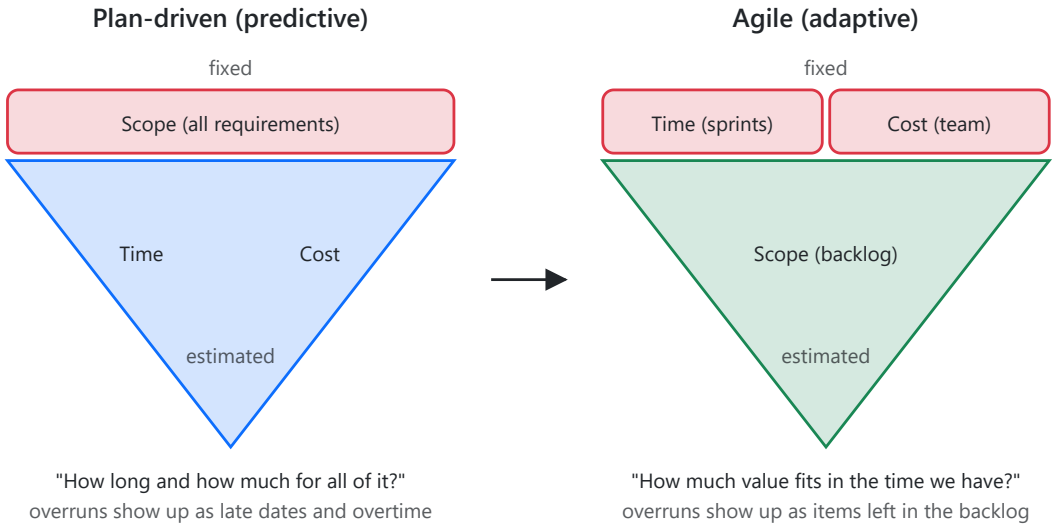
Baseline	Realtime Quiz artefact	Lab
Scope	Feature list with MoSCoW class, WBS	Lab-02, Lab-04
Schedule	Ten phases with dates, MVP checkpoint 2026-11-16	Lab-05
Cost	504 h planned by phase, 200 USD cash by month	Lab-06

- Sign convention here: **negative time variance = late**, **positive effort variance = over**. State your convention in the report header; earned value in Chapter 06 uses "negative is bad" for both.
- A variance report without a forecast and a proposed action is just bad news. The three actions keep the fixed sides fixed and move the one that may flex.

> Re-baselining is legitimate *after* approval. Silently editing the baseline to hide the slip is the mistake.

TOPIC 8

Agile turns the triangle upside down: fix time and cost, estimate scope



- The **Agile Practice Guide (2017)** describes the inversion: predictive approaches fix scope and estimate time and cost; adaptive approaches fix time (sprints) and cost (a stable team) and let scope vary.
- In Scrum (**Scrum Guide 2020**) the fixed box is the **Sprint**: one month or less, never extended. The Product Owner orders the backlog so the most valuable items are built first; what is not done stays in the backlog.
- Scope creep becomes harmless when the sprint is fixed: a new idea does not extend the sprint, it competes for a place in the next one.
- Quality is protected by the **Definition of Done**: an item is finished or it is not counted. No "90 % done".
- Realtime Quiz is exactly this shape: 14 weeks, 4 students, and a MoSCoW-ordered backlog.

Fixed sprints, measured velocity: how much scope fits before week 14?

```
Sprint length 2 weeks; team 4 x 10 h = 80 h per sprint (fixed cost)
7 sprints fit in 14 weeks; sprint 7 is the pilot (fixed time)

Sprint  Weeks  Capacity  Committed  Done   Note
S1      1-2    80 h      20 pts     18     setup, accounts
S2      3-4    80 h      21 pts     21     quiz authoring
S3      5-6    80 h      22 pts     17     WebSocket harder than planned

Velocity = average points done = (18 + 21 + 17) / 3 = 18.7 pts/sprint

Remaining backlog: 95 points (Must 30, Should 40, Could 25)
Sprints left before the pilot: S4, S5, S6 = 3 sprints
Forecast capacity: 3 x 18.7 = 56 points

Fits: Must 30 + top Should 26  ->  56 points
Does not fit: Should 14 + Could 25 = 39 points -> stay in the backlog

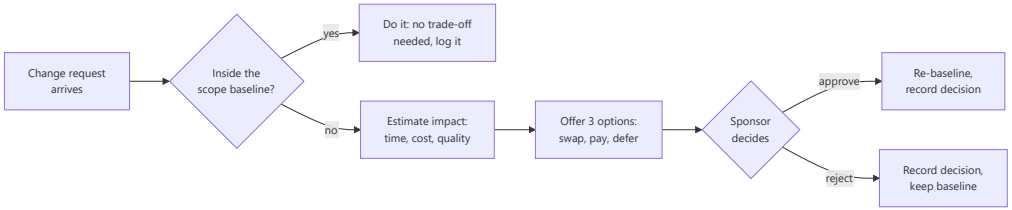
Decision for the Product Owner: order the 40 Should points so the
26 most valuable come first; tell the sponsor now, not in week 13.
```

	Predictive	Adaptive
Fixed	Scope	Time, cost
Estimated	Time, cost	Scope
Change	Change request, re-baseline	Reorder the backlog
Progress measure	Variance to baseline	Velocity, burndown
Bad news looks like	Late milestone	Items not in the release
Suits	Known requirements, contracts	Evolving requirements, fixed date

⚠ "Agile" is not an excuse to skip the trade-off: velocity is the data that makes the scope conversation honest. A team that hides velocity is back to three fixed sides.

TOPIC 9

Negotiating with sponsors: never "no", always "yes, and here is the price"




```
Sponsor: "Chat has to be in the pilot. Can you add it?"
PM:  "Yes, and here is what it costs. Chat is 40 hours. The
      team has 0 spare hours before week 14 and 56 hours of
      contingency reserved for the live-session phase.
      You can have chat if we defer the PDF export and video
      media (40 hours together), or if 480 USD buys a TA.
      Which do you prefer, or does chat stay in Could?"
Sponsor: "Defer the PDF export. Keep the pilot date."
PM:  "Agreed. I log CR-01 as approved with the swap and
      update the scope baseline today."
```

- **Lead with the fixed facts** the sponsor agreed to (the charter), not with your opinion.
- **Quantify** every option in the sponsor's units: weeks, USD, features at the pilot.
- **Offer choices, not a verdict**: the sponsor owns the trade-off, the PM owns the numbers.
- **Close with a record**: who decided what, when, and what was given up.

The decision record: one page that ends the argument for good

```
# DR-02 · Chat enters the Must scope, PDF export deferred

- Date: 2026-10-14 (week 3)      Status: approved
- Requested by: Head of IT (sponsor), change request CR-01
- Fixed constraints: pilot with 50 students in W14 (C-01),
  560 person-hours (C-02), 200 USD cash (C-03)

## Options considered
| Option | Time      | Cost     | Quality at pilot        |
|--------|-----------|----------|-------------------------|
| A slip | +1 week   | 0 USD    | breaks C-01             |
| B swap | +0 weeks  | 0 USD    | CSV export only         |
| C TA   | +0 weeks  | +480 USD | breaks C-03             |

## Decision
Option B. Chat (FR-N3) becomes Must. PDF export (FR-R4, PDF part)
and video media move to Could. Effort baseline stays 504 h.

## Consequences
Instructors get CSV only at the pilot. Chat needs a mute control
(abuse risk R-08). Deferred items are reviewed after the MVP demo.

Approved by: Head of IT · Recorded by: PM (lecturer)
```

Tactic sponsors use	Data-backed answer
"It is a small change."	"It is 12 hours. Here is the estimate and what 12 hours buys elsewhere."
"Just work a bit harder."	"Capacity is 40 h/week (C-02). Overtime in weeks 5-6 raised defects last time; here is the count."
"Other teams manage."	"Their charter had 6 people. Ours has 4; here are both."
"We can decide later."	"Deciding in week 10 costs 20 h of rework; deciding now costs 0. Deadline for the decision: W4."

A decision record is **Markdown in the repository**, next to the plan. The Lab-02 challenge writes DR-01 for a sponsor who wants every Should feature by week 10.

Case discussion: two projects with a fixed date, two different trades

	Windows "Longhorn" reset (2004)	HealthCare.gov launch (2013)
Situation	Three years into development the project was late and unstable; the flagship WinFS storage layer was not converging.	The federal insurance marketplace had a legal launch date of 1 October 2013 set by the Affordable Care Act.
Fixed side	Time: a shipping date that could not keep slipping, with a stable codebase as a quality floor.	Time: the date was written into law. Scope was treated as fixed too.
Trade made	**Scope cut**: the team restarted on the Windows Server 2003 codebase and dropped WinFS; the product shipped as Windows Vista (late 2006 / January 2007).	**No trade**: scope and date both held; testing was compressed into the final weeks. Quality gave way.
Outcome	A late but shipped product with a cut feature list; the dropped feature was never delivered.	The site failed under load at launch; a repair effort ("tech surge") followed and the site worked within about two months.
Lesson	Cutting scope explicitly is painful but visible and recoverable.	Refusing to trade does not protect scope; it moves the cut to quality, where nobody approved it.

- Both cases are publicly documented; the details above are simplified for discussion. Read the sources before quoting numbers.
- Question 1: in each case, **who decided** which side would give? Was it a decision or a default?
- Question 2: which **assumption** turned out false in each case (a storage layer can be finished in time; a site can be tested in a few weeks)?
- Question 3: what **data** would have made the trade-off visible earlier: velocity, defect counts, load test results?

⚠ The pattern repeats in student projects: a team that "will do everything" in week 12 ships an untested MVP in week 14. The trade happens either way; the only choice is whether it is deliberate.

If Realtime Quiz reaches week 10 behind schedule: what would you trade?

```
Scenario for discussion: status date end of week 10
- Must features done: 7 of 8 (reconnect-and-resume, FR-S9, open)
- Should done: responsive design only (1 of 5)
- Hours used: 420 of 560; contingency: 10 h left
- Pilot: week 14, 50 students, unchanged (C-01)

Candidate trades                  Saves    Risk to the pilot
1 Drop analytics charts, keep     24 h     instructors read CSV
  the CSV export
2 Drop notifications toasts       12 h     none for a pilot
3 Skip the load test with 100     16 h     HIGH: 50 real students,
  simulated students                       A-02 never validated
4 Move the pilot to week 15        -       breaks C-01: not allowed
5 Skip FR-S9 reconnect            20 h     HIGH: campus Wi-Fi drops,
                                           students lose scores

Defensible: 1 + 2 (36 h saved, quality intact).
Not defensible: 3 and 5 (they move the cut to quality).
```

Question for the group	What a good answer contains
Which side is fixed and why?	A quote from the charter (C-01 pilot in W14, C-02 team size), not a feeling.
What do we cut first?	Lowest MoSCoW class first, then the Should items with the smallest benefit per hour.
What may we never cut?	Anything in the Definition of Done and any Must feature; the load test validates assumption A-02.
How do we tell the sponsor?	A memo with three options priced in hours, and a decision deadline this week.
How do we prove it later?	A decision record in the repository and an updated scope baseline.

☑ This scenario is the raw material for the Lab-02 challenge: the sponsor asks for every Should feature by week 10 and you answer with velocity data and a counter-proposal.

WRAP-UP

Best practices and common mistakes

1 Name the driver in the charter

Write which side is fixed and which may flex before the first sprint. Realtime Quiz: fixed date and budget, scope flexes.

2 Price every change

No request enters the backlog without an estimate and a row in the change log. "Small" is not a number.

3 Keep Must at 60 % or less

The Should and Could classes are the buffer that makes a fixed date achievable. A 90 % Must list has no buffer.

4 Offer three options, never one

Swap, pay, or defer: each priced in time, cost and quality. The sponsor chooses; the PM keeps the arithmetic honest.

5 Validate assumptions early

Give each assumption an owner and a date before the phase that depends on it. Check WebSocket on campus in week 5, not week 14.

6 Record the decision

A decision record in Markdown, in the repository, with the options rejected and the consequences accepted.

Mistake: three fixed sides

Fixed scope, date and budget means quality is the flexible side, chosen by nobody. Name the trade or it names itself.

Mistake: crashing off the critical path

Money spent on a non-critical activity buys zero days. Check the path, crash the cheapest slope, re-check the path.

Mistake: editing the baseline

Moving milestone dates to hide a slip destroys the only measure you have. Re-baseline after approval, never silently.

Chapter quiz 10 questions

1. Weighted scoring: an option scores 4, 3 and 5 on criteria weighted 40, 35 and 25. What is its weighted score?

- 360
- 390
- 400
- 425

2. You must finish 2 days earlier. Activity A costs 200 USD per day saved (max 1 day), activity B costs 350 USD per day saved (max 3 days); both are critical. Minimum extra cost?

- 400 USD
- 550 USD
- 700 USD
- 750 USD

3. Effort baseline to the MVP checkpoint was 280 h; actual is 310 h. What is the effort variance in percent?

- +9.7 %
- +10.7 %
- +30 %
- -10.7 %

4. In the iron triangle, what sits in the centre and shrinks when the three sides are squeezed?

- Risk
- Benefits
- Quality
- Resources

5. The team adds a dark mode to the play page although no requirement or user asked for it. This is an example of:

- Scope creep
- Gold plating
- Fast-tracking
- Re-baselining

6. The Realtime Quiz charter fixes the pilot in week 14 and the team at 4 students. Which constraint must flex when the WebSocket phase runs late?

- Time
- Cost
- Scope
- None; work overtime

7. Planned scope was 80 feature points; by week 8 the work in progress represents 108 points. By what percentage has scope crept?

- 26 %
- 28 %
- 35 %
- 108 %

8. Which statement is an assumption rather than a constraint for the Realtime Quiz project?

- The pilot takes place in week 14.
- The campus network allows WebSocket connections.
- The cash budget is 200 USD.
- The stack is Spring Boot 4.1 with PostgreSQL.

9. According to the Agile Practice Guide, an adaptive (agile) approach fixes which sides of the triangle?

- Scope only
- Scope and time
- Time and cost
- Cost only

10. Starting the play page before the WebSocket API is final, so that two sequential activities overlap, is called:

- Crashing
- Fast-tracking
- Resource levelling
- Scope freezing

WRAP-UP

Summary

Scope, time and cost are the three levers; quality is the area between them and drops silently when the three are over-constrained.	Decide in the charter which side is fixed; Realtime Quiz is fixed-date and fixed-budget, so scope flexes through Should and Could.	Constraints are imposed and fixed; assumptions are believed and must be validated by an owner before a date.
Extended models add risk, resources and benefits; a project on time and budget that delivers no benefit has still failed.	MoSCoW keeps Must at or below 60 % of effort; weighted scoring = sum(weight × score), weights agreed before scoring.	Baselines change only by approved change; variance = actual - baseline, reported with a trend, a forecast and an action.
Scope creep and gold plating are unpriced scope; a change log with estimates and decisions makes them visible.	Crashing buys time with money (crash the cheapest cost per day on the critical path first); fast-tracking buys time with risk.	Agile fixes time and cost and lets scope vary; velocity is the data behind "yes, and here is the price", recorded in a decision record.

Open Lab-02: 5 tasks + 1 challenge

Next chapter: 03 · The project life cycle and methodology